

PHYSICS

PAPER 1

(THEORY)

Maximum Marks: 70

Time Allotted: Three Hours

Reading Time: Additional Fifteen Minutes

Instructions to Candidates

1. You are allowed **an additional fifteen minutes** for **only** reading the question paper.
2. You must **NOT** start writing during reading time.
3. This question paper has **11 printed pages**.
4. It has **twenty questions** in this paper. Answer **all** questions.
5. There are **four** sections in the paper: **A, B, C and D**. **Internal choices** have been provided in **two questions** each in **Sections B, C and D**.
6. **Section A** consists of one question having fourteen subparts of **one mark** each.
7. While attempting **Multiple Choice Questions** in Section A, you are required to **write only ONE option as the answer**.
8. **Section B** consists of **seven questions** of **two marks** each.
9. **Section C** consists of **nine questions** of **three marks** each.
10. **Section D** consists of **three questions** of **five marks** each.
11. The intended marks for questions are given in brackets [].
12. A list of useful constants and relations is given at the end of this paper.
13. A simple scientific calculator without a programmable memory may be used for calculations.

Instruction to Supervising Examiner

Kindly read **aloud** the Instructions given above to all the candidates present in the examination hall.

Note: The Specimen Question Paper in the subject provides a realistic format of the Board Examination Question Paper and should be used as a practice tool. The questions for the Board Examination can be set from any part of the syllabus, though the format of the Board Examination Question Paper will remain the same as that of the Specimen Question Paper. The weightage allocated to various topics, as given in the syllabus, will be strictly adhered to.

SECTION A– 14 MARKS

Question 1

(A) In questions (i) to (vii), choose the correct alternative (a), (b), (c) or (d) for each of the questions given below.

(i) A point charge $-Q$ and three points A, B and C are shown in **Figure 1** below. [1]



Figure 1

If V is electric potential and E is electric field intensity, we can infer that:

(Analyse)

(a) $E_A > E_B > E_C$

(b) $E_A < E_B < E_C$

(c) $V_A < V_C < V_B$

(d) $V_A > V_C > V_B$

(ii) Select the **odd** one out from the following options.

(Analyse) [1]

(a) Iron

(b) Nickel

(c) Cobalt

(d) Platinum

(iii) In **Figure 2** below, CD is an infinitely long and straight metallic wire carrying a current I . Point P is at a perpendicular distance r from it. [1]

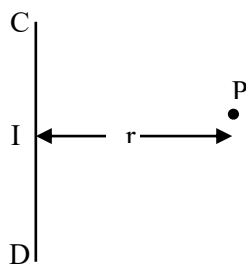


Figure 2

Which combination of I and r gives **maximum** magnetic field at point P, pointing into the **plane of the paper**?

(Analyse)

(a) $I = 4\text{A}$ towards C, with $r = 0.500\text{ m}$

(b) $I = 4\text{A}$ towards D, with $r = 0.250\text{ m}$

(c) $I = 8\text{A}$ towards C, with $r = 0.125\text{ m}$

(d) $I = 8\text{A}$ towards C, with $r = 0.050\text{ m}$

- (iv) Match the entries in Column I with those in Column II. (Recall) [1]

Column I	Column II
1. Planck	A. Hydrogen spectrum
2. Einstein	B. Wave nature of particles
3. Bohr	C. Quantum theory
4. de Broglie	D. Mass energy relation

- (a) 1-C, 2-D, 3-A, 4-B
(b) 1-B, 2-D, 3-C, 4-A
(c) 1-C, 2-A, 3-B, 4-D
(d) 1-A, 2-B, 3-D, 4-C
- (v) Identify the phenomena that occur when a P-N junction diode is forward biased. (Understand) [1]

- (P) The width of the depletion layer decreases.
(Q) The height of the potential barrier is lowered.
(R) The effective resistance of the diode increases significantly.
(S) The reverse current flows mainly due to the diffusion of majority charge carriers.

- (a) Both (P) and (Q)
(b) Both (R) and (S)
(c) Both (P) and (S)
(d) Both (Q) and (R)
- (vi) In an **astronomical telescope** of refracting type, (Recall) [1]
- (a) Objective lens has smaller focal length, and eyepiece lens has greater focal length.
(b) Objective lens has greater focal length, and eyepiece lens has smaller focal length.
(c) Both objective lens and eyepiece lens have large focal length.
(d) Both objective lens and eyepiece lens have small focal length.

- (vii) Given below are two statements marked, Assertion and Reason. Read the two statements and choose the correct option. [1]

Assertion: The focal length of a convex mirror will increase, if the mirror is placed in water.

Reason: The focal length of a convex mirror is equal to half its radius of curvature. (Analyse)

- (a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.
(b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.
(c) Assertion is true and Reason is false.
(d) Assertion is false and Reason is true.

- (B) Answer the following questions briefly:
- Two point charges Q_1 and Q_2 , separated by a distance, are such that there is no neutral point (where electric field is zero) in their vicinity. What is the relation between Q_1 and Q_2 ? [1]
(Recall)
 - Compare resistance of an ideal voltmeter with that of an ideal ammeter. [1]
(Understand)
 - How can the focal length of a given convex lens be measured in Physics laboratory using only a metre scale? [1]
 - What is the **difference** between a moving metallic ball colliding with another ball in motion and a photon colliding with an electron? [1]
(Understand)
 - When can a gamma ray photon produce a proton and an antiproton pair? [1]
(Recall)
 - Explain why resistance of a semiconductor material decreases when its temperature is increased. [1]
(Understand)
 - Fill in the blank with an appropriate term from the bracket.
When a light wave travels from one medium to another, its _____
(amplitude, wavelength, frequency, speed) remains unaltered. [1]
(Recall)

SECTION B – 14 MARKS

Question 2

[2]

- A negatively charged thundercloud above the earth's surface may be perceived as a parallel plate capacitor. (Refer to the image below.)
The lower plate of the capacitor is the Earth's surface, and the upper plate is the base of the thundercloud. The area of the base of the thundercloud $= 4.5 \times 10^8 \text{ m}^2$, distance of thundercloud base from the earth's surface $= 4425 \text{ m}$. Calculate the capacitance of this capacitor. (Apply)

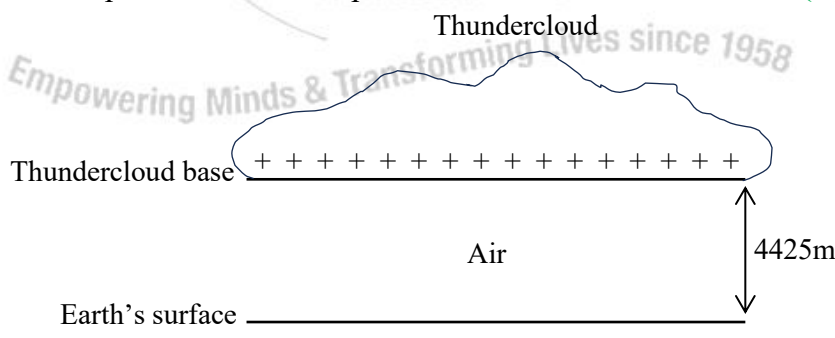


Figure 3

OR

- A tiny bird carrying a charge of $q = 8 \times 10^{-17} \text{ C}$ dives **vertically** down towards a large **positively** charged metallic plate kept on a table in a **horizontal** position. What should be the **surface charge density (σ)** of the plate so that the bird stops in its flight, close to the plate? (Mass of the bird $= 50 \text{ g}$) (Apply)

Question 3 [2]

Draw a labelled diagram of an arrangement used in Physics laboratory to compare emfs of two given cells X and Y. (Understand)

Question 4 [2]

- (i) According to Ohm's law, $V = IR$, so if potential difference (V) between the two ends of a conductor is varied, will its resistance (R) change? Explain your answer in brief. (Understand)

OR

- (ii) Heating power developed in a resistor R is given by $P = \frac{V^2}{R}$. It implies that power $P \propto \frac{1}{R}$ i.e., power varies inversely with R .
Heating power developed in a resistor R is also given by $P = I^2R$. It implies that power $P \propto R$ i.e., power varies directly with R .
Under what conditions, are the above-mentioned statements valid?

(Understand)

Question 5 [2]

Explain why a blue coloured spark appears in a switch contact when a circuit containing a load like a heater, geyser or an electric iron is switched off.

(Understand)

Question 6 [2]

There are seven types of electromagnetic waves having different frequency ranges. Arrange them in the increasing order of their frequencies. (Recall)

Question 7 [2]

- (i) Why the current flowing is extremely small during reverse biasing of a junction diode? (Understand)
- (ii) Extrinsic semiconductors are neither good nor bad conductors of electricity. Why are they still considered useful? (Understand)

Question 8 [2]

Two particles A and B of masses m_A and m_B have de Broglie wavelength λ_A and λ_B . If $\lambda_A = 2 \lambda_B$, compare their kinetic energies. (Analyse)

SECTION C – 27 MARKS

Question 9 [3]

- (i) Obtain an expression of intensity of electric field at a point lying on the axis of an electric dipole. (Understand)

OR

- (ii) An electric dipole is kept in a uniform electric field E with its axis making an angle θ with the field. Show forces acting on each charge and hence prove that torque acting is given by

(Understand)

$$\tau = pE \sin\theta$$

Question 10

[3]

- (i) Three identical cells each of emf 1.5V and internal resistance of 2Ω are connected in series to form a battery B. This battery is connected to an ammeter A (of negligible resistance) and five other resistors as shown in **Figure 4** below. Calculate the reading of the ammeter A.

(Apply)

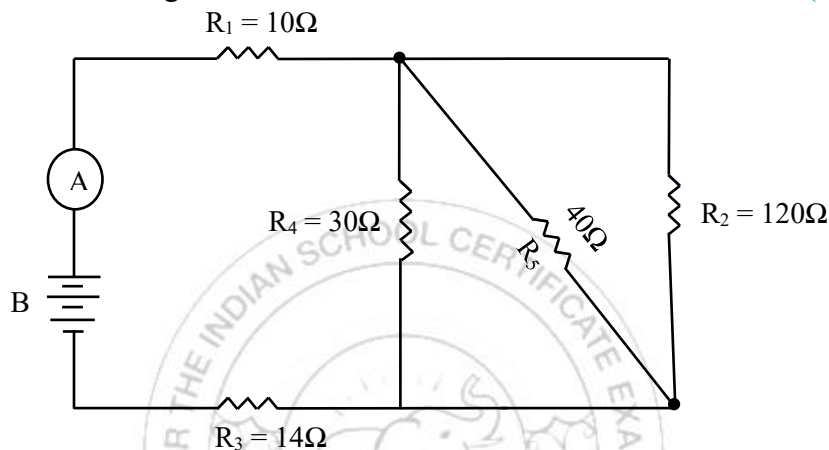


Figure 4

OR

- (ii) In the circuit shown in **Figure 5** below, the central zero galvanometer G shows no deflection when the key K is closed. Calculate the value of resistor R_1 .

(Apply)

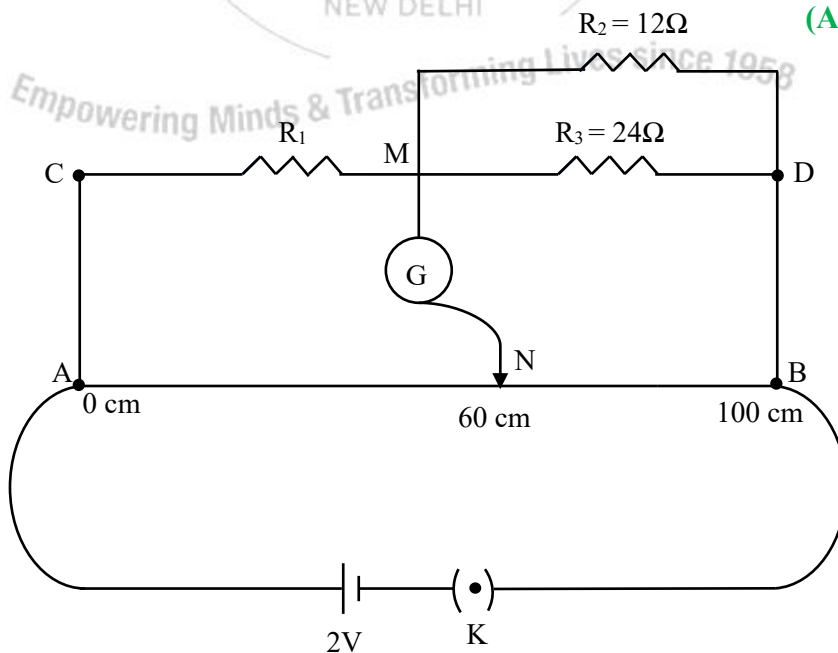
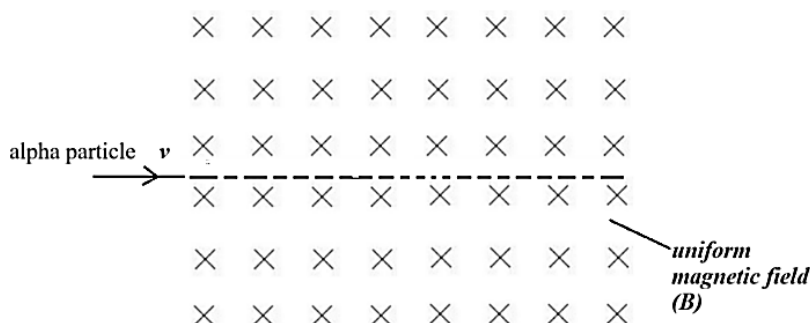


Figure 5

Question 11**[3]**

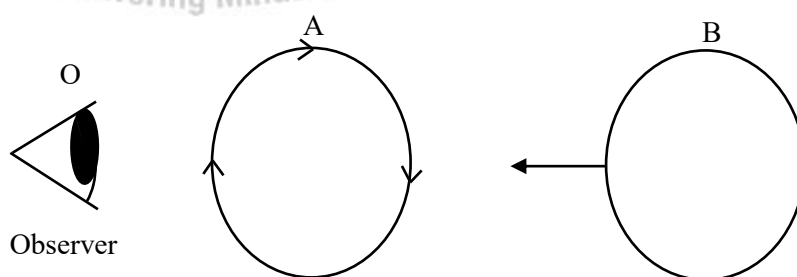
An alpha particle is moving with a velocity v . It enters a uniform magnetic field (B) as shown below. The magnetic field is perpendicular and into the plane of the paper. A uniform electric field is applied in the same region as the magnetic field so that the alpha particle passes undeviated through the combined fields.

**Figure 6**

- (i) What should be the direction of the electric field? **(Analyse)**
- (ii) Without any change in the electric and magnetic field, the alpha particle is replaced by a proton moving with a velocity v . Will there be any change in the path of the particle? Give a reason for your answer. **(Evaluate)**

Question 12**[3]**

- (i) State **any one** difference between the **nature** of magnetic field inside a current carrying solenoid and that along the axis of a current carrying circular coil. **(Recall)**
- (ii) A current is passed through a circular coil A in a **clockwise** direction, as shown in **Figure 7**. B is another circular coil which does not carry any current. However, when it is moved towards the coil A, current is induced in it. What is the direction of this induced current? **(Analyse)**

**Figure 7**

- (iii) Why is steel preferred to soft iron in making permanent magnets. **(Recall)**

Question 13**[3]**

A physics teacher drew a ray diagram on the board as shown below and asked the following questions to the students. State whether the answer of the student is correct or incorrect. Correct the incorrect response with an explanation in brief. **(Evaluate)**

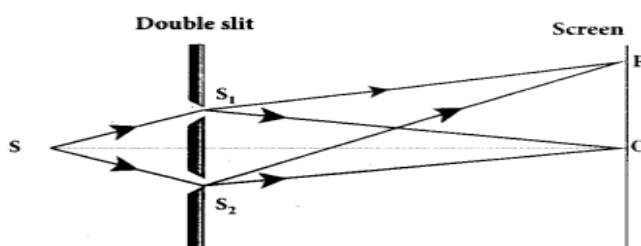


Figure 8

- (i) Teacher: There is a point P on the screen where the path difference between two interfering light waves is exactly 1.2×10^{-6} m. If the light emitted by the source has a wavelength of 600 nm, can anyone tell me whether a bright or a dark fringe will form at the point P?

Student: Dark fringe will be obtained.

- (ii) Teacher: Two students performed the same **Young's double slit experiment**. One student obtained wider fringes. What is the possible reason for this observation?

Student: The student who obtained wider fringes must have increased the distance between the two slits.

- (iii) Teacher: If one of the slits is closed with a finger while performing the experiment, what change will be noticed on the screen?

Student: The screen will get illuminated uniformly.

Question 14**[3]**

A student uses two prisms, A and B, made of different materials but having the same prism angle (60°).

Prism	Minimum Deviation
A	34°
B	47°

- (i) Calculate the refractive index of the material of prism A. **(Apply)**
- (ii) 'Since prism B has larger minimum deviation, the student claims it must also have the greater dispersive power.' Is this statement always correct? Justify your answer. **(Evaluate)**

Question 15**[3]**

- (i) An equiconvex lens forms a full size magnified image of a candle flame on a screen. How will this image change, if at all, when the lens is cut horizontally into two equal parts and only one of them is used? Explain. **(Analyse)**
- (ii) State what happens to magnifying power of a simple microscope if its focal length is decreased. **(Recall)**

Question 16**[3]**

Study the images shown in **Figure 9** below given below carefully and answer the questions that follow. (All three surfaces have same work function = 2.3 eV)

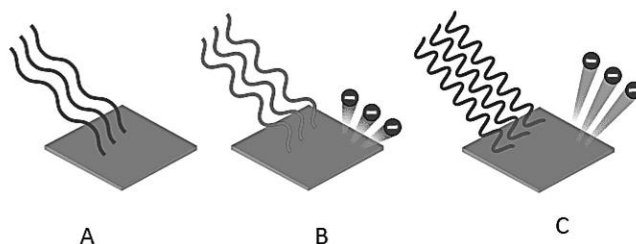


Figure 9

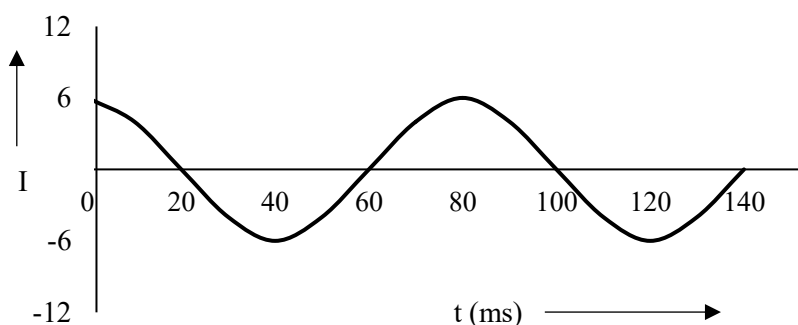
- Why are no photoelectrons emitted in image 'A'? **(Analyse)**
- Which image corresponds to the highest stopping potential? **(Recall)**
- If the photon energy in image B = 2.8 eV , calculate the **maximum kinetic energy** of the emitted photo electrons. **(Apply)**

Question 17**[3]**

- What is a **solar cell**? **(Recall)**
- Explain its working in brief. **(Recall)**
- In an energy band diagram of a certain material, forbidden band is absent. Identify the material. **(Recall)**

SECTION D – 15 MARKS**Question 18****[5]**

- (a) The graph below shows variation of current (I) flowing through an electrical device with time (t), when an ac source $e = 240\sin(\omega t)\text{ V}$ is connected to it.



- Identify the device. (resistor, capacitor or an inductor) **(Recall)**
- Calculate the frequency of the current flowing through the device. **(Apply)**
- How much is the opposition offered by this device? **(Apply)**

- (b) A train is moving on horizontal tracks that are 1.2m apart. Its displacement (x) depends on time (t) as $x(m) = 3t^2 + 6$. Vertical component of earth's magnetic field in its region of motion is 2×10^{-5} T. Calculate the emf induced between two ends of its axle rod 10s after the train starts moving. **(Apply)**

OR

- (ii) An ac generator generating an emf $e = 300 \sin(100\pi t)$ V is connected to a 100Ω resistor.
- Draw a **labelled** graph of current (I) flowing through it versus time (t). Draw one cycle only. **(Recall)**
 - Calculate the **average power** consumed by the resistor. **(Apply)**
 - Why ac is preferred to dc, though it has varying values whereas the latter has a fixed value? **(Understand)**
 - State **any one** difference between a transformer used at the power generating station and the transformer in a city which is used to supply power to households. **(Recall)**

Question 19

[5]

- Write **Bohr's** equations for circular motion of an electron around the nucleus and conservation of angular momentum. Using them, obtain an expression for **orbital velocity** of an electron. **(Apply)**
 - Prove that $1u = 931.5$ MeV where the terms have their usual meaning. **(Apply)**

OR

- Assuming that nuclei are spherical in shape, whose radius R is given by $R = R_0 A^{1/3}$ where R_0 is a constant, show that the nuclear density (i.e. density of nuclear matter) is independent of mass number (A) of the nucleus / atom. **(Apply)**
 - What are isotones? **(Recall)**
 - Name the series of lines in hydrogen spectrum which lies in **(Recall)**
 - Visible region**
 - UV region**

Question 20

[5]

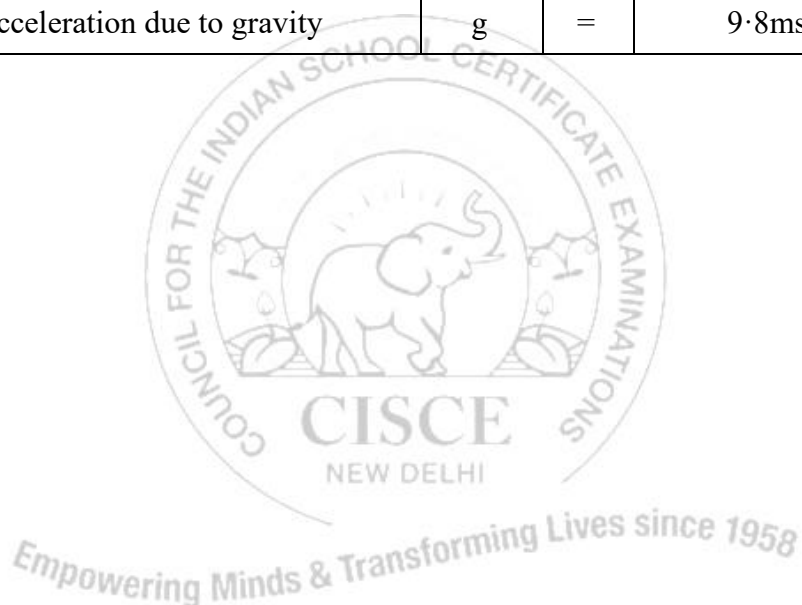
An innovative optical sensor system for a miniature drone incorporates a combination of various optical elements. A beam of light from an LED source first passes through a convex spherical refracting surface (radius of curvature $R = 20$ cm) separating air $\mu_1 = 1.0$ from a glass medium of refractive index $\mu_2 = 1.5$. The light then enters a thin biconvex lens of focal length 10 cm. Finally, the beam strikes a concave spherical mirror of radius of curvature 30 cm, which acts as a collector, focusing the rays onto a photodetector (optical instrument application) to measure light intensity.

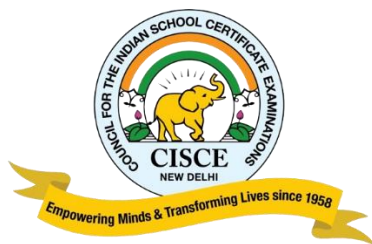
- If the drone operates in a highly humid or misty environment where a film of water completely coats the lens surfaces, how will the refractive power of this lens change? **(Apply)**

- (ii) A point object is placed in air at a distance of 60 cm in front of the convex spherical refracting surface. What is the position of the image formed by this surface? **(Understand)**
- (iii) If this setup is to be modified to function as a reflecting astronomical telescope, what roles would be played by the mirror and the lens? **(Understand)**
- (iv) Which mirror collects light from a distant object most efficiently? **(Recall)**

USEFUL CONSTANTS AND RELATIONS

1.	Mass of an electron	m	=	$9.1 \times 10^{-31} \text{ kg}$
2.	Permittivity of vacuum	ϵ_0	=	$8.85 \times 10^{-12} \text{ Fm}^{-1}$
3.		1u	=	$1.66 \times 10^{-27} \text{ kg}$
4.	One electron volt	1 eV	=	$1.6 \times 10^{-19} \text{ J}$
5.	Acceleration due to gravity	g	=	9.8 ms^{-2}





PHYSICS
PAPER 1
(THEORY)
ANSWER KEY

(Equivalent answers are acceptable.)

SECTION A– 14 MARKS

Question 1

(A) In answering Multiple Choice Questions, candidates have to write either the correct option number or the statement against it. Please note that only ONE correct answer should be written.

- (i) (c) $V_A < V_C < V_B$ [1]
- (ii) (d) Platinum [1]
- (iii) (d) $I = 8\text{A}$ towards C, with $r = 0.050\text{ m}$ [1]
- (iv) (a) 1-C, 2-D, 3-A, 4-B [1]
- (v) (a) Both (P) and (Q) [1]
- (vi) (b) Objective lens has greater focal length, and eyepiece lens has smaller focal length. [1]
- (vii) (d) Assertion is false and Reason is true. [1]

(B)

- (i) $Q_2 = -Q_1$ Or Q_1 and Q_2 form an electric dipole. [1]
- (ii) $R_v = \infty$ and $R_a = 0$ [1]
Voltmeter has a very high resistance while an ammeter has a very low resistance.
- (iii) The focal length of a convex lens can be measured using only a metre scale through the Distant Object Method. This method relies on focusing light rays from a very distant source onto a surface to find the lens's focal point. [1]
- (iv) When a photon collides with an electron, it loses either its **entire** energy or **none** at all, whereas a moving ball can lose **any fraction** of its energy. [1]
- (v) The gamma ray photon should have the minimum energy of $2(m_p)c^2$. [1]
Or
The gamma ray photon should have the Combined Minimum Energy
 $E = 938.3 + 938.3 = 1876.6\text{MeV}$.
- (vi) As temperature rises, large number of electrons break their covalent bonds and cross the forbidden band gap into the conduction band. Thus, there are large number of free electrons in the conduction band which result in reduction in resistance. [1]
- (vii) Frequency [1]

SECTION B – 14 MARKS

Question 2

[2]

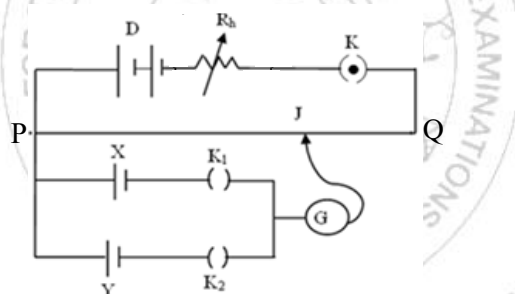
$$\begin{aligned}
 \text{(i)} \quad C &= \frac{\epsilon_0 A}{d} \\
 &= \frac{8.85 \times 10^{-12} \times 4.5 \times 10^8}{4425} \\
 &= 0.009 \times 10^{-4} \\
 &= 9 \times 10^{-7} \text{F}
 \end{aligned}$$

OR

$$\begin{aligned}
 \text{(ii)} \quad qE &= mg \\
 q \frac{\sigma}{2\epsilon_0} &= mg \\
 \sigma &= \frac{2mg\epsilon_0}{q} \\
 &= \frac{2 \times 50 \times 10^{-3} \times 9.8 \times 8.85 \times 10^{-12}}{8 \times 10^{-17}} \\
 &= 10.8 \times 10^4 \text{Vm}^{-1}
 \end{aligned}$$

Question 3

[2]



Question 4

[2]

- (i) No, resistance R of a conductor is constant. Its value does not depend on V or I .

OR

- (ii) $P \propto \frac{1}{R}$ when V is constant i.e. when resistors are connected in parallel.
 $P \propto R$ when I is constant i.e., when resistors are connected in series.

Question 5

[2]

This is due to a large induced emf which produces an electric discharge through air. Ionised air molecules emit light when they return to ground state from their excited states.

Question 6**[2]**

The correct arrangement of the seven electromagnetic waves in increasing order of their frequencies (from lowest to highest frequency) is:

1. Radio Waves (Lowest frequency)
2. Microwaves
3. Infrared Waves
4. Visible Light
5. Ultraviolet (UV) Rays
6. X-rays
7. Gamma Rays (Highest frequency)

Question 7**[2]**

- (i) The current flowing through the diode during reverse bias is extremely small because it is entirely due to minority charge carriers, which are much smaller in number (in comparison with number of majority carriers).
- (ii) Extrinsic semiconductors are highly useful because their electrical conductivity can be precisely controlled, tuned, and manipulated by altering the concentration of dopant atoms (impurities) added to the crystal.

Or

Electronic devices like diodes, transistors, solar cells etc. can be made out of extrinsic semiconductors, which have many applications in electronic gadgets.

Question 8**[2]**

$$\frac{(KE)_A}{(KE)_B} = \frac{\frac{h^2}{2m_A\lambda_A^2}}{\frac{h^2}{2m_B\lambda_B^2}}$$

$$\frac{(KE)_A}{(KE)_B} = \frac{m_B\lambda_B^2}{m_A\lambda_A^2}$$

$$\text{Given: } \lambda_A = 2\lambda_B$$

Now,

$$\frac{(KE)_A}{(KE)_B} = \frac{m_B\lambda_B^2}{m_A(2\lambda_B)^2}$$

$$\frac{(KE)_A}{(KE)_B} = \frac{m_B}{m_A \times 4}$$

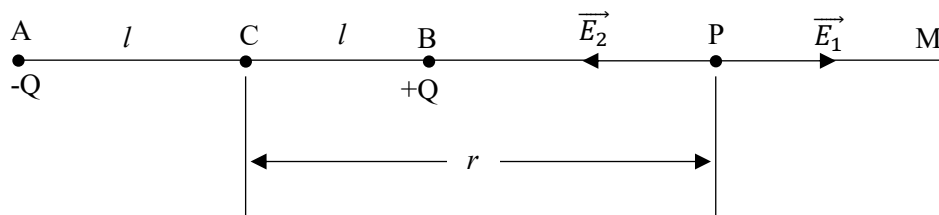
$$(KE)_A : (KE)_B = m_B : 4m_A$$

SECTION C – 27 MARKS

Question 9

[3]

(i)



$$E_1 = K \cdot \frac{Q}{(r-l)^2}$$

$$E_2 = K \cdot \frac{Q}{(r+l)^2}$$

$$E = E_1 - E_2$$

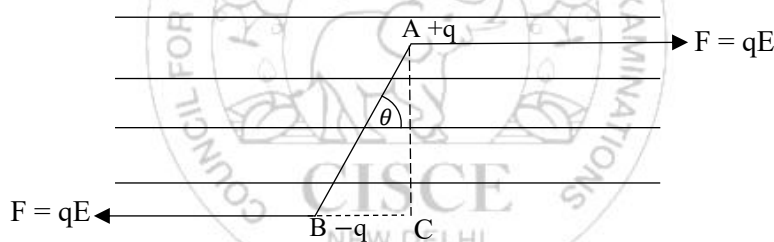
$$= K Q \left[\frac{1}{(r-l)^2} - \frac{1}{(r+l)^2} \right]$$

$$E = K \cdot \frac{2pr}{(r^2-l^2)^2}$$

This is the required expression.

OR

(ii)



$$\tau = \text{Force} \times \text{distance}$$

$$= qE \times AC$$

$$= qE \times \left(\frac{AC}{AB} \right) \times AB$$

$$= qE \sin \theta \times L$$

$$\tau = pE \sin \theta \text{ (where, } p = qL \text{)}$$

This is the required expression.

Question 10

[3]

(i) R_4 , R_5 and R_2 are in parallel.

Their equivalent resistance R_6 is given by

$$\frac{1}{R_6} = \frac{1}{30} + \frac{1}{40} + \frac{1}{120}$$

$$\therefore R_6 = 15\Omega$$

Now, R_1 , R_6 and R_3 are in series.

$$\therefore R = 10 + 15 + 14 = 39\Omega$$

Current I delivered by the battery is given by

$$I = \frac{E_{\text{Total}}}{R+r} = \frac{4.5}{39+6}$$

$$= 0.1\text{A}$$

Reading of ammeter = 0.1A

OR

- (ii) R_2 and R_3 are in parallel.

Their equivalent resistance R_4 is given by

$$\frac{1}{R_4} = \frac{1}{12} + \frac{1}{24}$$

$$\therefore R_4 = \frac{24}{3} = 8\Omega$$

It's a balanced Wheatstone bridge

So,

$$\frac{R_1}{R_4} = \frac{l}{(100-l)}$$

$$\frac{R_1}{8} = \frac{60}{40}$$

$$\therefore R_1 = \frac{60 \times 8}{40} = 12\Omega$$

Question 11

[3]

- (i) $\vec{E}$ should be in the plane of the paper, pointing vertically downwards.

- (ii) Since alpha particles travel undeviated;

$$F_m = F_e$$

$$Bqv = qe$$

$$\therefore v = \frac{E}{B}$$

Since protons also obey the abovesaid relation (same v , same E and same B), they will also travel undeviated.

Question 12

[3]

- (i) Magnetic field inside a solenoid is **uniform** whereas magnetic field along the axis of a circular coil is **non uniform**.
- (ii) In anticlockwise direction (according to Lenz's law)
- (iii) It has a large coercive force i.e. it is more difficult to demagnetise steel than iron.

Question 13

[3]

- (i) Incorrect, bright fringe (since path difference = 2λ)
- (ii) Incorrect, because the slit separation should have been decreased.
- (iii) Incorrect, diffraction pattern is seen on the screen.

Question 14**[3]**

- (i)
$$n = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \frac{\sin\left(\frac{60+34}{2}\right)}{\sin\left(\frac{60}{2}\right)} = 1.47$$
- (ii) The student's statement is not always correct. A larger minimum deviation indicates a higher mean refractive index, whereas dispersive power depends on the difference in refractive indices for violet and red light.

Question 15**[3]**

- (i) The nature, size, and position of the image will not change at all, but the brightness (intensity) of the image will be reduced to half because there is no change in focal length and same object and image distances.
- (ii) Magnifying power increases when the focal length is decreased.

Question 16**[3]**

- (i) No photoelectrons are emitted because the energy of the incident photons is less than the work function of the material. For photoelectric emission to take place, the incident photon energy must be greater than or equal to the threshold work function.
- (ii) C
- (iii) Using Einstein's photoelectric equation:
 $K_{\max} = E - \phi$
 $K_{\max} = 2.8 - 2.3 = 0.5 \text{ eV}$
 The maximum kinetic energy of the emitted photoelectrons in image B is 0.5 eV.

Question 17**[3]**

- (i) A solar cell (also known as a photovoltaic cell) is a semiconductor p-n junction diode that directly converts solar energy (light energy) into electrical energy.
- (ii) The incoming photons break covalent bonds near the junction, generating free electron-hole pairs. This build-up of charges creates a voltage difference (electromotive force). When an external load or circuit is connected, a direct electric current begins to flow.
- (iii) Metal

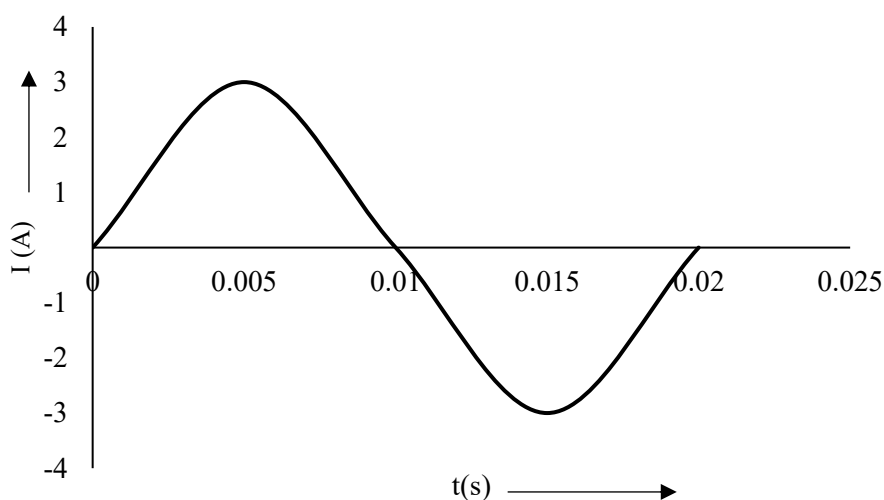
SECTION D – 15 MARKS**Question 18****[5]**

- (i) (a) (1) Capacitor (current is leading over applied emf by 90°)
- (2) $f = \frac{1}{T} = \frac{1}{80 \times 10^{-3}} = 12.5 \text{ Hz}$
- (3) $X_c = \frac{e_0}{i_0} = \frac{240}{6} = 40 \Omega$

(b) $x = 3t^2 + 6$
 $\therefore \frac{dx}{dt} = 6t$
 $\therefore v = 6t$
 After 10s, $v = 6 \times 10 = 60\text{ms}^{-1}$
 $E = BLv$
 $= 2 \times 10^{-5} \times 1.2 \times 60$
 $= 1.44 \times 10^{-3} \text{ V}$

OR

(ii) (a) $i_0 = \frac{e_0}{R} = \frac{300}{100} = 3.00\text{A},$
 $2\pi f = 100\pi$
 $f = 50 \text{ Hz}$
 $T = \frac{1}{f} = \frac{1}{50} = 0.02\text{Hz}$



(b) $\langle P \rangle = \frac{I_0^2 R}{2}$
 $= \frac{9 \times 100}{2}$
 $= 450\text{W}$

- (c) Because ac can be stepped up or down with the help of a transformer whereas dc cannot be stepped up or down.
 (d) The former is a step up transformer whereas the latter is a step down transformer.

Question 19

[5]

(i) (a) $\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{(Ze)(e)}{r^2}$
 $mvr = n \frac{h}{2\pi}$
 $mv^2 = \frac{Ze^2}{4\pi\epsilon_0} \frac{(2\pi mv)}{nh}$
 $\therefore v = \frac{Ze^2}{2\epsilon_0 nh}$

(b) $1u = 1.66 \times 10^{-27} \text{ kg}$

$$\begin{aligned}
 c &= 3 \times 10^8 \text{ m/s} \\
 E &= mc^2 \\
 E &= 1.66 \times 10^{-27} \times (3 \times 10^8)^2 \\
 &= 14.929 \times 10^{-11} \text{ J} \\
 E &= \frac{14.929 \times 10^{-11}}{1.6 \times 10^{-19}} = 9.315 \times 10^8 \text{ eV} \\
 &= 931.5 \text{ MeV}
 \end{aligned}$$

OR

- (ii) (a) $R = R_0 A^{1/3}$
 where,
 $R_0 = \text{constant}$
 $A = \text{mass number of the nucleus}$
 Nuclear matter density, $\rho = \frac{\text{mass of the nucleus}}{\text{volume of the nucleus}}$
 Let m be the average mass of the nucleon.
 Hence, mass of the nucleus = mA
 $\therefore \rho = \frac{mA}{\frac{4}{3}\pi R^3}$
 $= \frac{3mA}{4\pi \left(R_0 A^{1/3}\right)^3}$
 $= \frac{3mA}{4\pi R_0^3 A} = \frac{3m}{4\pi R_0^3}$
- (b) Isotones are atoms or nuclei that have the same number of neutrons (N) but different numbers of protons (Z)
- (c) (1) Balmer series lies in the visible region.
 (2) The Lyman series lies in the ultraviolet (UV) region.

Question 20

[5]

- (i) The refractive power of the lens will decrease (focal length will increase).
- (ii) $\frac{1.5}{v} - \frac{1}{-60} = \frac{1.5-1}{20}$ solving this $v = +180 \text{ cm}$
 The image is formed at a distance of 180cm inside the denser medium (glass), measured from the pole of the refracting surface.
- (iii) Objective: The concave spherical mirror must act as the objective.
 Eyepiece: The thin biconvex lens must act as the eyepiece.
- (iv) A parabolic mirror